SkyPass: Exact Weather-Aware Pass Scheduling with Learned Sky and Brightness Models

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***Abstract*—Observation planning at a small ground station is distributed across unconnected tools: one predicts passes, another evaluates optical visibility, and cloud and conflicts between overlapping passes are left to the operator. This paper presents SkyPass, an integrated planner that carries orbital elements through propagation, pass extraction, visibility scoring, cloud-forecast calibration and exact conflict resolution to an executable timetable, with two optional learned components. Pass extraction agrees with an independent implementation to 0.055 s while requiring 23.7× fewer propagator evaluations than a dense scan. Conflict resolution under a nightly observing budget is solved exactly by a dynamic program linear in the number of candidate passes and the nightly limit; it matched exhaustive search on all 2000 randomised instances. Forecast skill is verified against ERA5 reanalysis at seven stations and scheduling value at six, over 60 days. Forecast value is determined by the decision it informs: a station required to observe every night obtains no benefit, whereas one able to select its nights improves realised yield by 19.0% (expected value) and 37.6% (clear-sky count), and calibration raises the gain in every regime. Learned segmentation reads the cloud fraction of a whole-sky camera image to within 1.2 percentage points, against 3.5 for a tuned colour-ratio rule, and gradient-boosted trees trained on 17 216 measured magnitudes reduce the brightness error for unseen satellites from 1.51 to 0.83 mag, an advantage that does not persist into a later period with new satellite designs.**

***Index Terms*—satellite pass prediction, ground station scheduling, interval scheduling, forecast calibration, cloud segmentation, machine learning**

I. Introduction

Small ground stations, such as university laboratories, amateur-radio clubs and individual optical observers, face a recurring planning problem. Several hundred satellites are potentially observable and over a week generate thousands of passes, of which only a few dozen merit the operator's time, and the station can attend to one object at a time.

Existing tools address only parts of this problem. Pass predictors compute rise and set times from two-line elements (TLEs) and the SGP4 propagator [1], [2], and some add an illumination test so that optical observers are not shown daylight passes. A few applications display a weather indication beside a pass, but to our knowledge none allow the forecast to *rank* passes or drive a scheduling decision, and none resolve conflicts between overlapping passes.

Weather is the more consequential omission. At the running example (Chennai, India, 12.92° N) mean observed cloud over optically viable passes was 88%, yet a predictor ranking by elevation still recommends an overcast night. Once several satellites are tracked their passes also overlap, and a single-aperture station must select among them; the scheduling literature treats this from the operator side, as spacecraft tasking and range scheduling [3]–[7], with constraints and solvers a small station does not need.

The contributions are: an integrated pipeline accounting for the passes removed by each physical constraint; a validated pass extractor; exact capacity-constrained conflict resolution requiring neither a solver nor a metaheuristic; a weather claim verified against ERA5 reanalysis [8]; two negative results, namely that weather-awareness yields no benefit under a fixed nightly quota and that the uncalibrated forecast degrades performance there; a measured element-set error budget over 33 223 aged comparisons; and two learned components, a cloud nowcast read from whole-sky images and a data-driven photometric model, each evaluated against classical baselines.

II. Related Work

SGP4 and its two-line element sets remain the standard basis for rapid satellite position prediction [1], [2]. Accuracy is limited by the element sets rather than the propagator: TLE-derived positions carry errors that grow with element-set age, as measured against on-board GPS [9].

Whether a satellite is *seen* depends on illumination and photometry, not only geometry. Standard treatments use a conical Earth-shadow model [1] and a diffuse-sphere phase function for apparent magnitude [10], and megaconstellation brightness work has refined these [11]. Measured magnitudes are now collected in bulk by the IAU CPS SCORE repository [12], which makes a learned photometric model possible.

Satellite range scheduling is an oversubscribed combinatorial problem [3], usually approached with multi-objective optimisation, metaheuristics or reinforcement learning [5]. Earth observation scheduling is a large adjacent literature [7], [13], in which cloud is an uncertainty hedged during imaging tasking [14], [15], always on the *operator* side. Weighted

interval scheduling, the single-machine case, is a textbook dynamic program [16].

Professional observatories schedule around weather using queue schedulers that re-plan as conditions change [17], [18], but assume an institutional software stack and on-site sky monitoring. Forecast verification and forecast value are mature fields [19], and statistical post-processing is essential before a forecast drives a decision [20]. Sky imaging estimates cloud by thresholding the red-to-blue ratio [21] or, more recently, with convolutional segmentation [22]. To our knowledge neither has been applied to observer-side pass planning.

III. System Design

SkyPass is a five-stage pipeline (Fig. 1). Each stage is a pure function of the preceding one, so any stage may be ablated independently, which is the mechanism by which the weather claim is evaluated.

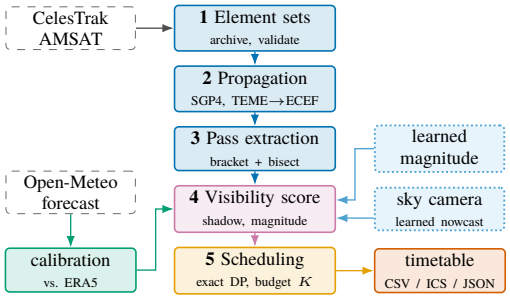


Fig. 1. SkyPass pipeline. The cloud forecast is calibrated against reanalysis before it is allowed to influence the score; the scheduler then resolves conflicts exactly under the station's observing budget. Dotted: optional learned components.

A. Element sets, propagation and pass extraction

Element sets are fetched from CelesTrak [23], checksum-validated, deduplicated per NORAD identifier keeping the newest epoch, and *archived under the date of retrieval*: CelesTrak serves only current elements, so without a local archive a published prediction cannot be repeated. Sets whose epoch age exceeds $\tau_{\max}$ are rejected.

Positions come from the reference SGP4 implementation. The TEME vector is rotated to ECEF by Greenwich mean sidereal time and converted to topocentric azimuth, elevation and range on the WGS-84 ellipsoid. Polar motion is neglected, being below 15 m at the surface and far under the SGP4 error budget. Atmospheric refraction [24] is reported but not applied to the horizon mask, so results do not depend on assumed surface conditions.

Because cost is dominated by propagator calls, pass extraction is designed to minimise them. Elevation is sampled on a coarse grid to *bracket* horizon crossings; within a pass elevation is smooth and unimodal, so a sign change of $\varepsilon(t) - \varepsilon_{\text{mask}}$ between adjacent samples brackets exactly one crossing. Each bracket is refined by bisection to 0.1 s, and culmination by golden-section search. The coarse step is $P/240$ bounded to 5 s to 30 s, so short-period orbits, whose passes are shortest, are sampled most finely.

B. Visibility scoring

Each pass receives a score $S \in [0, 1]$ combining a geometric quality term with multiplicative gates:

$$S = \underbrace{(w_e f_e + w_d f_d)}_{\text{geometry}} \cdot f_m \cdot f_s, \tag{1}$$

where $f_e = \min(\varepsilon_{\max}/60°, 1)$ and $f_d = \min(T/480\,\text{s}, 1)$ saturate culmination elevation and duration, f_m is a photometric factor and f_s a sky-condition factor.

The multiplicative form is deliberate: an eclipsed pass or one in a bright sky scores zero irrespective of its geometry, and expected cloud scales the score down in proportion, which an additive score cannot represent. For optical work the object must be sunlit and the observer in darkness; Earth shadow uses the dual-cone construction [1], and apparent magnitude follows the standard diffuse-sphere relation [10], [25]

$$m = m_{\text{std}} + 5\log_{10}\frac{d}{1000\,\text{km}} - 2.5\log_{10} F(\psi), \tag{2}$$

with $F(\psi) \propto (\pi - \psi)\cos\psi + \sin\psi$ normalised to $F(0) = 1$. In radio mode the illumination and photometric terms are dropped.

C. Weather integration and calibration

The sky factor is $f_s = 1 - \hat{c}$, where $\hat{c}$ is the expected cloud fraction at culmination, interpolated linearly between hourly samples. Crucially $\hat{c}$ is *not* the raw forecast, which is heavily over-confident as a predictor. We fit

$$\hat{c} = \mathbb{E}[c \mid c_f] = a + b\,c_f \tag{3}$$

by least squares on a window that *ends before* the period being planned, restricted to local night hours, and clip $\hat{c}$ to $[0, 1]$. The slope b quantifies the strength of the signal: $b \approx 1$ indicates that the forecast may be used directly, whereas $b \approx 0$ indicates that it carries little information, in which case the calibrated value converges toward site climatology, the appropriate fallback. For the threshold value model the planner instead weights each pass by the calibrated probability of clear sky, $P(c \le \theta \mid c_f)$, estimated in ten forecast bins on the same window and shrunk toward the site's base rate with 20 pseudo-observations, so each value model is paired with the planner that maximises its expectation.

D. Conflict resolution

A single-aperture station observes one object at a time and requires a setup gap g between targets, so selecting the best mutually compatible subset of passes is weighted interval scheduling. Writing $v_j = S_j \pi_j$ for the value of pass j (score times mission priority), indexing passes $1, \ldots, n$ by finish time and letting $p(j)$ be the largest $i < j$ compatible with j (0 if none), the textbook recurrence [16] gives the exact optimum in $O(n \log n)$.

This formulation alone is insufficient: left unconstrained, a station observes every candidate, so weather-awareness has no selection to perform. Operational stations are limited by operator and instrument time, so we impose a capacity budget

of K observations per night, extending the recurrence with a cardinality dimension:

$$\mathrm{OPT}[j][k] = \max\big(\mathrm{OPT}[j-1][k],\ v_j + \mathrm{OPT}[p(j)][k-1]\big), \quad (4)$$

with $\mathrm{OPT}[0][k] = \mathrm{OPT}[j][0] = 0$. This is exact. The table has one entry for each of the n candidate passes and each count up to the nightly limit K, and each entry is filled in constant time, so after the sort the cost is $O(nK)$: doubling either the candidates or the limit doubles the work. Observing nights are disjoint in time, so applying the budget per night and solving each night independently remains exact. Nights are defined in local solar time so that an evening and the small hours that follow belong to the same session.

E. Learned components

Two optional models replace hand-built approximations. The hourly forecast cannot discriminate within a night (Section V-D); a whole-sky camera can. Two segmenters label each pixel of a camera image as cloud or sky: a U-Net [26] with three down-sampling stages, trained with a cross-entropy masked to the fisheye and with rotations and flips, since the sky has no preferred azimuth; and gradient-boosted trees on fifteen per-pixel colour and texture features. The cloud fraction of the image replaces $\hat{c}$ for the next hour. For photometry, gradient-boosted regression trees [27] predict the apparent magnitude from the phase angle, range, elevation, solar elevation and orbital altitude at culmination, the prediction of (2), the satellite family and the observing band. The family is known for every catalogued object, so the model applies to satellites that have never been measured.

IV. Experimental Setup

The catalogue comprises 635 unique objects from CelesTrak GP groups, checksum-validated and deduplicated per object. Seven stations span distinct cloud climatologies: Chennai and Bengaluru (monsoon), Leh (high-altitude arid), Tucson (desert), Singapore (equatorial), and Reykjavík and Svalbard (maritime polar). Unless stated otherwise, the mask is 10°, the dark-sky threshold −6°, and the setup gap 5 min.

All three weather products come from Open-Meteo [28]: the operational forecast; archived past forecasts at lead times of 1–7 days; and ERA5 reanalysis [8] as verification truth. The archive endpoint has to be pinned to the ERA5 model: its default best-match series *is*, for recent dates, the archived forecast, and verifying against it reports a spurious skill of 1.0.

Timing is checked against a brute-force 1 s dense scan and against Skyfield [29], which shares only the certified SGP4 core and implements time scales, frames and event finding independently. All runs are single-threaded, under Python 3.13.5 with `sgp4` 2.27 on a commodity laptop.

Measured magnitudes come from SCORE [12]: after removing flagged, unilluminated and below-horizon rows and the solar sail ACS 3, 17 216 observations of 2709 satellites remain, cross-validated in five folds grouped by satellite and split in time at 2025-07. Sky images are the 400 labelled images of WSISEG [22], cross-validated in five contiguous blocks.

TABLE I
Pass-timing agreement over the full catalogue (635 objects, 1761 matched passes, 24 h horizon). Values in seconds.

	vs. 1 s dense scan		vs. Skyfield	
Quantity	mean	p95	mean	p95
$\lvert\Delta t_{\mathrm{AOS}}\rvert$	0.504	0.954	0.060	0.157
$\lvert\Delta t_{\mathrm{TCA}}\rvert$	0.248	0.476	0.055	0.110
$\lvert\Delta t_{\mathrm{LOS}}\rvert$	0.502	0.956	0.059	0.156
$\lvert\Delta\varepsilon_{\max}\rvert$ (deg)	0.0008	0.0022	0.0034	0.0102

Baselines see identical data: the red-to-blue ratio rule [21] at 0.6 and tuned, and the sphere with tabulated or per-family fitted standard magnitudes.

V. Results

A. Pass-prediction accuracy and cost

Table I compares SkyPass against both references over 635 objects and 1761 matched passes.

Against Skyfield, mean agreement is 0.060 s on acquisition and 0.055 s on culmination. The apparent 0.504 s "deviation" from the dense scan is not SkyPass error: a 1 s grid locates a crossing only to within its own step, so it lags the true crossing by a mean of 0.5 s. The refined estimator is thus more accurate than the brute-force reference it is conventionally compared against. The cost difference is large: 2 311 148 propagator evaluations against 54 864 000, a factor of 23.7. Detection recall is 99.78%, the residual being very short grazing transits that peak barely above the mask.

B. Element-set limits on prediction accuracy

Sub-second numerical agreement is meaningful only if the orbital elements support that precision. In the operational catalogue the median element-set age was 0.42 days (90.2% under one day). CelesTrak and AMSAT differ by a median of 0.00 s in culmination over 308 passes, bounding redistribution rather than orbit-determination differences.

To measure decay with staleness we obtained historical element sets for 60 low-Earth objects, took the freshest set before a target night as reference, and re-predicted the same passes from progressively older sets, giving 33 223 aged comparisons (Fig. 2).

Within a day of epoch the elements contribute 0.05 s of culmination error, the same order as the 0.055 s numerical agreement of Section V-A, so neither term dominates at that age. Degradation is then sharply super-linear and heavy-tailed: a mean of 4.3 s (median 0.70 s) at 7–14 days, 16.2 s at 14–30, and 75.3 s (median 13.69 s) beyond a month. The design consequence is that a 5–7 day staleness cutoff is appropriate for timing-critical work, beyond which the elements rather than the algorithm set achievable accuracy, and that daily element refresh is the single largest lever on prediction accuracy.

C. Skill of the cloud forecast

Weather-aware planning is justified only if the forecast is informative at the lead times over which an observer plans.

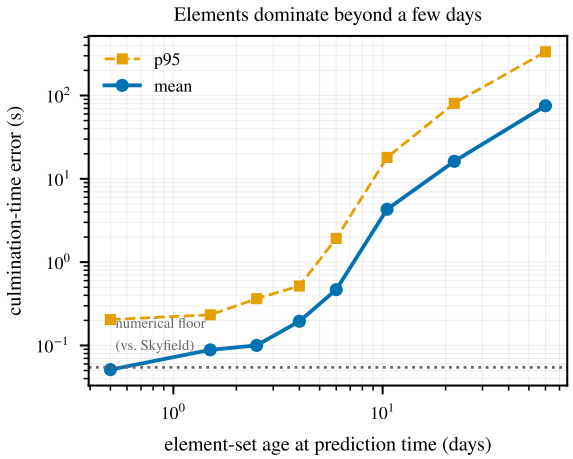


Fig. 2. Culmination-time error against element-set age, over 33 223 aged comparisons. The dotted line is the numerical agreement with Skyfield. Fresh elements land on that floor; by a week the mean is nearly two orders of magnitude above it.

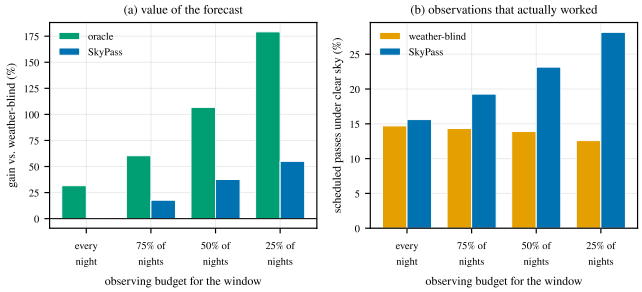


Fig. 3. Weather-awareness is worth nothing to a station obliged to observe every night, and a great deal to one that may choose its nights. (a) Gain over the weather-blind exact planner, threshold value model; the oracle plans with the observed cloud. (b) Share of scheduled passes under clear sky.

Archived forecasts were verified against ERA5 across all seven stations over 60 days, scoring the operational binary decision of whether a given hour is sufficiently clear to observe at a 30% cloud threshold. Accuracy alone is misleading when clear hours are rare, so we report the Heidke skill score, a chance-corrected measure in wide use for categorical forecast verification [30].

Skill decays steadily with lead time, from HSS 0.347 at one day to 0.289 at three days and 0.114 at seven. The one-day forecast outperforms both naive baselines (persistence 0.206, climatology 0 by construction), but by five to seven days it no longer does. A weather-aware planner should therefore run on a rolling short horizon and re-plan daily. Skill is also strongly site-dependent, the spread between stations being comparable to the decay with lead time and lowest at the two Indian monsoon stations, Chennai reaching zero skill beyond four days. The expected benefit of weather-awareness is thus a local property rather than a universal constant.

D. Value of weather-aware scheduling

Five planners select from an identical candidate set, drawn from 300 objects, at each of six stations over 60 days (Svalbard has no observable passes), and every timetable is scored against ERA5. Since the valuation of a clouded observation is a modelling choice, we report both an *expected value* model, $\sum_j v_j(1-c_j)$, and a *threshold* model, $\sum_j v_j \mathbb{1}[c_j \le \theta]$ with $\theta = 30\%$.

Under a fixed nightly quota the forecast yields no benefit: SkyPass gains -1.5% under the expected-value model and -0.2% under the threshold model, against oracle headroom of 12.0% and 31.6% respectively, and the station-bootstrap intervals of both gains include zero. This is not because the forecast is uninformative: at the candidate-pass timestamps, forecast and observed cloud correlate at $r = 0.41$, and the clearest-forecast decile does identify clearer skies, reducing observed cloud by 0.209.

The limitation is that a nightly quota permits the planner only to *reorder passes within a night*, which two measured properties render ineffective. Cloud is strongly autocorrelated over the few hours a session spans, the between-night standard deviation of observed cloud being $2.2\times$ the within-night value; and pass quality varies more than sky conditions do, the ninetieth-percentile pass carrying $1.8\times$ the mean weather-free value. Since the score is a product, selecting for clear sky requires surrendering base value, an exchange that is unfavourable where quality is the more variable factor.

Given the same nightly cap together with a budget for the window, so that it may *skip* a night, the result reverses (Fig. 3): the choice of which nights to use is a day-scale decision, the scale at which the forecast retains skill. Under the budget permitting observation on 50% of nights (90 observations across the window), SkyPass improves realised yield by 19.0% under the expected-value model and 37.6% under the threshold model (95% intervals 9.2–24.9% and 3.1–80.4%), recovering 41% and 35% of the attainable oracle gains. The fraction of scheduled passes falling under clear sky rises from 13.9% to 23.1%, and tightening the budget to a quarter of the nights raises the threshold-model gain to 55.0%.

Calibration raises the gain in every regime. Under the quota, supplying the model cloud field directly to the score costs -9.2% relative to disregarding weather; under the half budget the raw field still helps (8.0% and 29.9%) but less than the calibrated one. The fitted calibration indicates the cause: slopes lie between 0.17 and 0.61 and intercepts between 0.07 and 0.70, so clear forecasts are followed by more cloud than predicted and cloudy ones by less. Calibration (3) shrinks the forecast toward climatology in proportion to its demonstrated reliability.

With epoch-correct geometry (the element set current on each night) the threshold-model gain rises under the quota to 9.2% and at a quarter budget to 75.9%, but falls at the half budget to 33.2%, the expected-value gain there staying at 19.0%: back-propagated geometry does not inflate the result systematically.

E. Conflict resolution in practice

On 2000 randomised instances small enough to enumerate exhaustively, the dynamic program matched the exhaustive optimum in 100.00% of cases. Greedy selection *by the score itself* attains 99.95% of the optimum on average, so the dynamic program offers little average-case improvement over it; its advantage is that it is exact, has negligible cost, and exhibits no adverse case. The rules an operator typically applies perform markedly worse (Fig. 4), since they optimise a proxy rather than the objective: selection by highest culmination attains 95.39% on average, is optimal in only 62.7% of instances, and falls to 9.4% in the worst case generated.

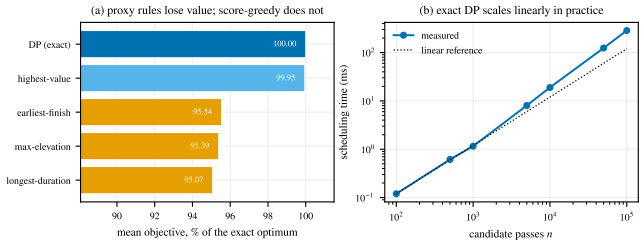


Fig. 4. (a) Greedy heuristics attain less than the exact dynamic program. (b) Scheduling time grows linearly in practice and remains negligible at catalogue scale.

On real candidate sets an order-based genetic algorithm (population 60, 120 generations) attains 99.67% of the optimum at about 38 922× the time, so a metaheuristic is unnecessary; the dynamic program handles 100 000 candidates in 285.3 ms.

F. End-to-end behaviour

The visibility funnel is steep and physically interpretable: at the demonstration site, 12 449 geometric passes reduce to 8093 sunlit, 1297 with the observer in darkness, 367 bright enough to detect, 87 above the score floor once forecast cloud is weighted in, and 26 scheduled. Svalbard (78.2° N) has the most geometric passes of any station yet *zero* observable ones, the Sun never falling below −6° there in late August.

Planning 635 objects over 7 days takes 109.4 s, dominated by propagation. Varying w_e from 0.3 to 0.7 leaves a Jaccard overlap of at least 0.89 with the default timetable, so the physical constraints, not the weighting, govern the outcome.

G. Learned components

The fixed colour-ratio threshold does not transfer to this camera, misestimating the image cloud fraction by 27.0 points on average; tuned on the training images it errs by 3.5. Both learned segmenters reduce the error to 1.2 points (Fig. 5), with cloud intersection-over-union 0.945 for the pixel classifier and 0.927 for the U-Net. The classifier wins every fold and is deployed, needing no deep-learning framework. All images come from one camera.

For satellites absent from training the learned photometry errs by 0.83 mag, against 1.51 for the sphere with fitted per-family standard magnitudes and 1.86 with tabulated ones;

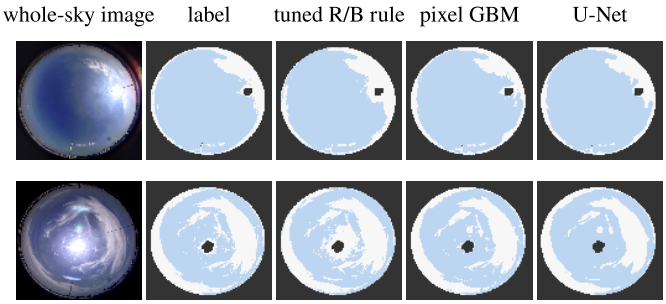


Fig. 5. Held-out whole-sky images, their labels, and the predictions of the tuned colour-ratio rule and the two learned segmenters (cloud white, sky blue, undefined dark).

orbital altitude and family carry most of its skill. Trained before 2025-07 and tested after, however, it ties the fitted sphere (2.00 against 1.99 mag): better on the 6 new families, worse on established ones as designs change. The sphere stays the default.

VI. Discussion and Limitations

Cloud cover is an imperfect proxy for observing conditions: total cloud fraction on a reanalysis grid cell of order 25 km describes a line of sight only coarsely, and transparency, seeing, cloud height and aerosol are not modelled. A camera nowcast can be read from an image with small error (Section V-G), but its planning value could not be measured, since no public camera archive covers the evaluation nights. Neither value model captures partial credit; the scheduler accepts a station's own value function unchanged.

The gains pool six stations and 60 nights in one season. Their 95% intervals come from a bootstrap that resamples whole stations (10 000 draws); with so few clusters they are approximate and probably too narrow, and the calibrated-minus-raw difference was not tested separately.

We assume one instrument observing one object at a time; multiple apertures or a station network break the interval-scheduling structure and would require a different algorithm. The learned photometry narrows the gap between the sphere's ordering and a prediction only while its training data describe the satellites in orbit. The ageing measurement is single-season and restricted to low Earth orbit, where drag-driven decay dominates.

VII. Conclusion

SkyPass integrates pass prediction, optical visibility analysis, cloud-forecast reasoning and exact conflict resolution into a single planner suitable for deployment at a small ground station, producing a timetable that exports directly to a calendar or a rotator controller. Pass extraction agrees with an independent implementation to 0.055 s while requiring 23.7× fewer propagator calls than dense sampling, and conflict resolution is exact at negligible computational cost.

Two weather-related findings are of particular relevance to the design of such systems. First, a forecast does not by itself improve performance: it is informative about which *nights*

merit observation and much less so for ranking passes within a night, where cloud is autocorrelated and pass quality varies substantially more than sky conditions. A planner under a fixed nightly quota cannot exploit that information and gains nothing, whereas given a budget it may allocate selectively the same forecast yields 19.0% (expected value) and 37.6% (clear-sky count). Second, calibration against reanalysis raises the gain in every regime tested, and under a quota it is the difference between a loss and break-even. Both findings should carry over to other small facilities, though only one setting is tested here: the value of a forecast is a property of the decision it informs. The learned components extend the same principle: segmentation of a whole-sky image supplies the sub-hourly signal a quota-bound station lacks, and learned photometry predicts magnitudes for unseen satellites far better than the sphere, but only while its training data describe the satellites in orbit. The implementation and experiment scripts are released under the MIT licence.

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